

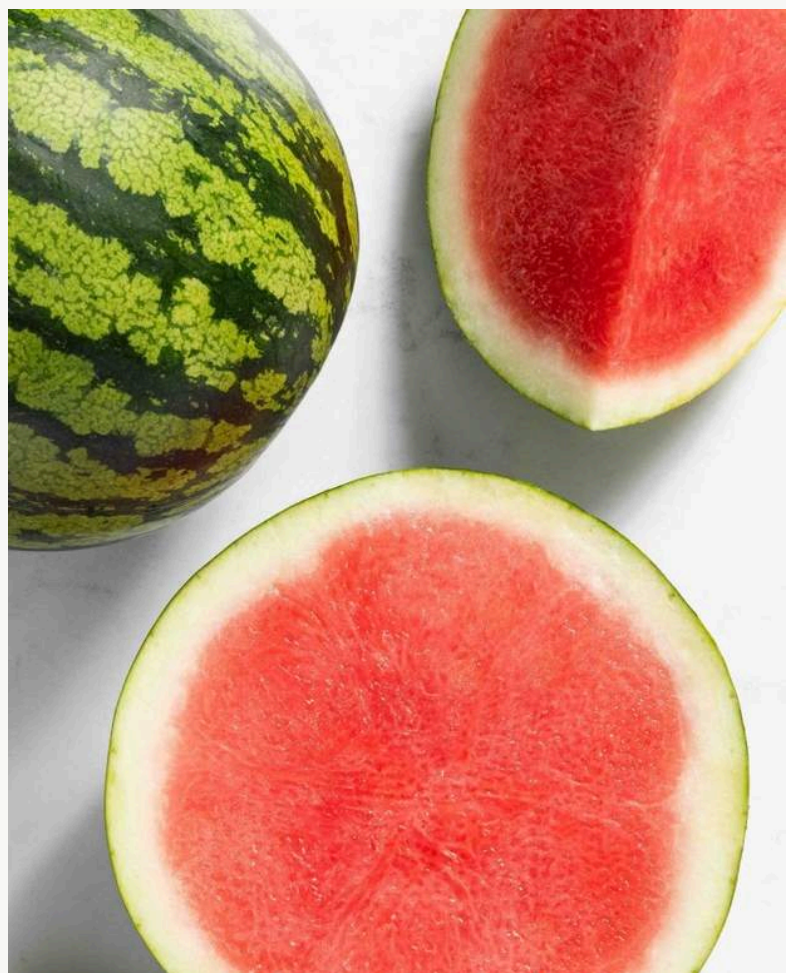
The 2026 Watermelon Internal Quality Benchmark

How a few sampled fruit can have an outsized impact on a 40,000 lb shipment

A field-based reference for U.S. watermelon growers, packers, and shippers — for sizing the economic exposure created by internal-quality uncertainty.

INSIDE

- 01 The benchmark and the sampling gap
- 02 What the exposure costs
- 03 Where the exposure sits



A shipment judged by a handful of fruit

A buyer may inspect and cut only a handful of fruit before deciding whether a shipment meets expectations. At the packing line, destructive sampling and the know-how of a few experienced tappers are the only line of defense. **Most of the fruit remains unseen inside.**

3.5%

AVERAGE REJECTION
RATE

1%

TARGETED REJECTION
RATE

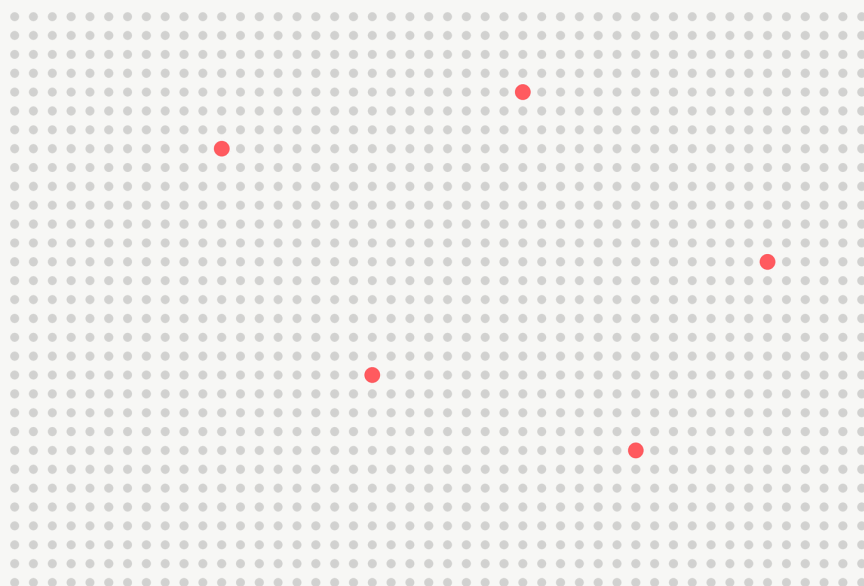
\$300K

REPRESENTATIVE ANNUAL
EXPOSURE, 50-75M LB

40K lb

AVERAGE SHIPMENT SIZE

The 3.5% figure is a practical field reference, **not a statistically representative national average**. Strong producers may already perform materially better — and should be able to demonstrate it.



ILLUSTRATIVE. ONE SHIPMENT HOLDS THOUSANDS OF FRUIT; THE DECISION MAY REST ON A HANDFUL.

The sampling gap

Sampling is necessary — cutting every watermelon open is impossible. But internal conditions may not be distributed evenly, the result depends on which fruit are selected, and sampled fruit can no longer be sold as packed product.

The largest cost is not the fruit destroyed. It's the commercial decision made with incomplete information.

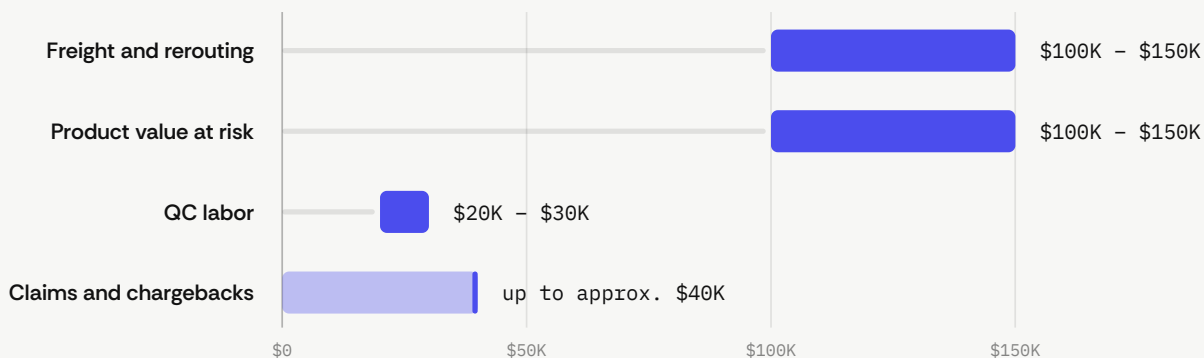
● SAMPLED ● UNSEEN

A rejection rate is an economic measure

Rejection is often discussed as a percentage. The consequences arrive as dollars — freight already spent, additional rerouting expense, a lower price after diversion or downgrade, a supported claim, and time spent coordinating another destination.

Not every rejection creates every cost, and not every affected shipment becomes a total loss. A credible benchmark reflects the mix of outcomes rather than applying the worst case to every event.

REPRESENTATIVE COMPOSITE MODEL · 50-75 MILLION LB SEASONAL OPERATION



\$300,000

APPROXIMATE REPRESENTATIVE ANNUAL ADDRESSABLE EXPOSURE

This is not a promise that every operation will save \$300,000. It is a realistic reference point for deciding whether the problem deserves a closer look. Components are shown as ranges and are not additive to a point total. Four of the five sources are quantified here; sampled-fruit loss is operation-specific.

WHY THE APPROXIMATE 1% TARGET MATTERS

Moving from the 3.5% reference to the approximate 1% target closes a gross operating gap of **25 shipments per 1,000**. What that gap is worth depends on three questions.

01 How many events are connected to internal quality?

02 What happens after each rejection: acceptance, claim, discount, downgrade, or reroute?

03 Could better information have changed grading or routing before shipment?

Rejection rate alone is not enough. Producers need rejection incidence and economic severity.

The approximately \$300,000 reference excludes destructive-sampling savings, premium pricing, incremental revenue, Orbem pricing, and unverified consequences such as shelf-life deterioration, replacement obligations, disposal loss, or future buyer allocation.

Where the exposure actually sits

Five places to put a number against your own operation. The answer should be an operation-specific range — not a universal promise.

01

Freight and rerouting

Freight already spent on an affected load, plus the added cost of moving it somewhere else.

02

Product value at risk

Value per pound lost when a load is diverted or downgraded, and what earlier routing could preserve.

03

QC labor

Shift time absorbed by sampling and subjective internal-quality calls.

04

Destructively sampled fruit

Fruit cut open in a season, at packed value. Operation-specific, so excluded from the \$300,000.

05

Claims and chargebacks

Documented claims tied to internal quality, net of value already counted under 02.

START WITH TWO NUMBERS — THE FIVE SOURCES ABOVE ASK FOR THE REST

01

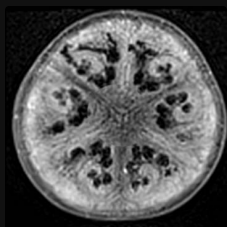
Volume and loads

Annual pounds, shipment count, and average payload.

02

Rejection incidence and reasons

Current rate, by buyer and destination — and what drove each event.



FROM A FEW SAMPLES TO EVERY-FRUIT INFORMATION

Orbem's Genus Echo inspects internal watermelon quality non-destructively at up to 44,000 lb/hour.

The difference is coverage. Sampling uses a few fruit to represent a shipment; every-fruit inspection can provide internal-quality information before fruit reaches the buyer — informing grading, holding, releasing, routing, field and variety analysis, and buyer-specific quality management.

BENCHMARK YOUR OPERATION

REQUEST A CONFIDENTIAL INTERNAL-QUALITY REVIEW

BENCHMARK
CONFIDENTIAL REVIEW

amkappes-cmd.github.io/orbem-watermelon
calendar.app.google/1pP3A1nvstA7CGHr8

METHODOLOGY AND LIMITATIONS — Based on detailed anonymized work with two U.S. watermelon producers, site observations, a representative composite economic model, and broader field conversations. Not a statistically representative national survey. It does not assume that every rejection is caused by internal quality, that every

event produces a total-load loss, or that every affected shipment can be prevented. Results vary with volume, rejection definition, buyer practices, shipping distance, product value, recovery channel, staffing, season length, and the share of exposure connected to internal quality.